Model Alpha M14 Synthetic Sampling Methods Study

HELIX FORGE LAB / p04-full-001070 / RUN R184 / 2026-07-13

ALPHA F1	PRECISION	RECALL	BETA F1
0.873	0.881	0.865	0.846

Decision context

This report records a bounded synthetic sampling evaluation on HF-Eval-S1/S2 v3.2. The 2,400-record population is synthetic. Alpha M-14 is retained as a research checkpoint, not as a deployment authorization.

Engineering interpretation

Aggregate improvement remains meaningful only with the frozen scoring policy. The review therefore keeps slice-level calibration, dataset lineage, and the comparison with Beta B-07 attached to run R184.

Lineage

Upstream evidence: p04-full-000149 (research/programs/model-alpha/experiments/results/2026-07-13/model-alpha-m14/reproducibility/specifications/model-alpha-m14-reprodu

Slice	Records	Precision	Recall	F1
slice-01	174	0.871	0.872	0.871
slice-02	251	0.888	0.848	0.868
slice-03	166	0.870	0.861	0.865
slice-04	189	0.887	0.874	0.880
slice-05	219	0.869	0.850	0.859